

Make-Up Water Systems: Process, Flow & Optimization Overview Purpose:

Make-up water systems replenish losses in HVAC loops, boilers, chillers, and cooling towers. They ensure system pressure and volume remain stable, while conditioning incoming water to prevent scale, corrosion, and biofouling.

System Start-to-Finish Flow:

1. **City/Well Water Supply Entry**

- Water enters through a backflow preventer to prevent contamination.
- Isolation and check valves control flow path.

2. **Pretreatment**

- Water passes through softeners to remove hardness (Ca, Mg).
- May include filters, carbon beds, or dechlorinators depending on application.

3. **Flow Control & Regulation**

- Float-operated or conductivity-based control valves regulate when to feed make-up water.
- Booster pumps ensure proper delivery pressure.

4. **Injection to System Loop**

- Treated water flows into surge tank, feedwater tank, or tower basin.
- Flow meters log make-up rate for water balance tracking.

Key Components & Definitions:

- ****Backflow Preventer****: Prevents reverse flow of contaminated water.
- ****Water Softener****: Removes hardness minerals to prevent scale.
- ****Make-Up Valve****: Opens when system level or conductivity drops.
- ****Flow Meter****: Measures gallons added per day/week.
- ****Booster Pump****: Maintains pressure in high-rise or large systems.

Example Flow:

> A chilled water loop in a 10-story building uses a conductivity probe in the expansion tank to signal when make-up water is needed. Upon drop in conductivity (indicating volume loss), a solenoid valve opens to allow pre-softened city water into the system. Flow is tracked with a digital flow meter (GPM/day).